Discrete Ricci Flow On Special Graph Structures

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Notations and Definitions

The following definitions are from "Ollivier Ricci-flow on weighted graphs" by Shuliang Bai, Yong Lin, Linyuan Lu, Zhiyu Wang, and Shing-Tung Yau.

Let $G = (V, E, w)$ be a weighted graph. The function $w((x, y)) : E \to [0, \infty)$ is the associated edge weight between vertices $(x, y) \in E$. We denote $x \sim y$ as vertices that have an edge weight between them, and w_{xy} to be the edge weight. The distance metric $d(x, y)$ is the length of a minimal weighted path among all paths that connect vertices x, y.

Definition 1. A probability distribution over the vertex set $V(G)$ is a mapping $\mu : V \to [0, 1]$ defined by

$$\mu_x^\alpha(y) = \begin{cases} \alpha & \text{if } y = x \\ (1-\alpha)\frac{\gamma(w_{x,y})}{\sum\limits_{z\sim x}\gamma(w_{x,z})} & \text{if } y \sim x \\ 0 & \text{otherwise} \end{cases}$$

Where $\gamma : \mathbb{R}_+ \to \mathbb{R}_+$ represents an arbitrary one-to-one function, and $\alpha \in [0, 1]$ controls the amount of mass left at node x.

Definition 2. A coupling between two probability distributions μ_1 and μ_2 is a mapping $A : V \times V \to [0, 1]$ with finite support such that

$$\sum_{y\in V} A(x, y) = \mu_u(x) \quad \text{and} \quad \sum_{x\in V} A(x, y) = \mu_v(y)$$

Definition 3. The transportation distance between two probability distributions μ_1 and μ_2 is given by the Wasserstein Function

$$W(\mu_1, \mu_2) = \inf_A \sum_{x,y\in V} A(x, y)d(x, y)$$

Where the infimum is taken over all coupling A between μ_1 and μ_2.

The Wasserstein distance is also equal to the optimal solution to the dual problem

$$W(\mu_1, \mu_2) = \sup_f \sum_{x \in V} f(x)[\mu_1(x) - \mu_2(x)]$$

Where the supremum is taken over all 1-Lipschitz functions f. f is 1-Lipschitz if it satisfies

$$|f(x) - f(y)| \leq d_{x,y} \text{ for } \forall x, y \in V(G)$$

Definition 4. Let $G = (V, E, w)$ be a weighted graph where the distance metric d is determined by the weight function w by shortest weighted paths between nodes. For any $(x, y) \in E(G)$ and $\alpha \in [0, 1]$, the α-Ricci curvature k_α is defined to be

$$k_\alpha(x, y) = 1 - \frac{W(\mu_x^\alpha, \mu_y^\alpha)}{d(x, y)}$$

And the Lin-Lu-Yau adaptation of Ollivier Ricci curvature is defined as

$$k(x, y) = \lim_{\alpha \to 1} \frac{k_\alpha(x, y)}{1 - \alpha}$$

Definition 5. Given an initial metric $w_i(0)$ for each edge weight in graph G, the system of ordinary differential equations given by

$$\frac{dw_e(t)}{dt} = -k_e(t) w_e(t)$$

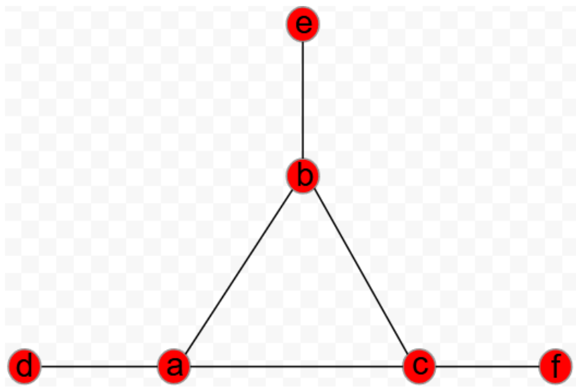
defines the unnormalized flow on graph G.

The normalized flow which ensures the sum of all edge weights remains constant 1 throughout time is given by

$$\frac{dw_e(t)}{dt} = -k_e(t) w_e(t) + w_e(t) \sum_{h \in E(G)} k_h w_h(t)$$

Definition 6. Let $G = (V, E)$ be equipped with an initial metric $w \in \mathbb{R}_{>0}^{|E|}$ and let the metric $w(t)$ evolve under the normalized Ricci flow. We say that the normalized Ricci flow converges to a metric with constant curvature τ if there exists a metric $w(\infty) \in \mathbb{R}_{>0}^{|E|}$ such that $\lim_{t \to \infty} w(t) = w(\infty)$ and under the metric $w(\infty)$ the curvature on each edge in $E^+ := \{e \in E(T) : w_e(\infty) > 0\}$ is τ.

Complete 3 Graph Plus 3 leaf edges



Define Graph $G = V, E$ With $V = \{a, b, c, d, e, f\}$ and $E = \{(a, b), (a, c), (b, c), (a, d), (b, e), (c, f)\}$. The structure is a complete K_3 graph plus one more edge from each node, like in the image. We will perform an analysis on the long time behavior of the edge weights in G under normalized Ollivier-Lin-Lu-Yau Ricci flow.

Initial Metric

Let the initial metric be all edge weights initially equal,

$$w_{ab}(0) = w_{ac}(0) = w_{bc}(0) = w_{ad}(0) = w_{be}(0) = w_{cf}(0)$$

Symmetry between edges

Since the initial metric has all equal edge weights, the symmetrical structure of the graph induces symmetry between internal edges and leaf edges. for all time,

$$w_{ab}(t) = w_{ac}(t) = w_{bc}(t) \quad \text{and} \quad w_{ad}(t) = w_{be}(t) = w_{cf}(t)$$

This results in only two types of edges. Let w_i refer to the three internal edges, and w_l refer to the three leaf edges. Then we have

$$w_i := w_{ab} = w_{bc} = w_{ac} \quad \text{and} \quad w_l := w_{ad} = w_{be} = w_{cf}$$

Distance Metric

For two nodes x and y where $(x, y) \in G(E)$ has a shortest distance that is not the edge weight w_{xy} at any point in time, then a distance violation is triggered. For leaf-internal edges like (a, d), the distance violation will never occur since any path must include the edge weight itself. For internal edges, a distance violation cannot occur since under the symmetrical metric, at $t = 0$ it is not violated and the internal edge weights will evolve symmetrically preserving this fact.

Hence for all time, the shortest distance between any two nodes that have an edge between them is the edge weight. From this it follows that

$$d(x,y) = \begin{cases} w_i & \text{if } (x,y) = (a,b),(a,c),(b,c) \\ w_l & \text{if } (x,y) = (a,d),(b,e),(c,f) \\ w_l + w_i & \text{if } (x,y) = (d,b),(d,c),(a,e),(c,e),(a,f),(b,f) \\ 2w_l + w_i & \text{if } (x,y) = (d,e),(d,f),(e,f) \end{cases}$$

Where $d(x,y)$ is the shortest distance between nodes x, y.

The mass distribution (**Definition 1.**) for nodes a, b, d are as follows

$$\mu_a(a) = \alpha \quad \mu_a(b) = \frac{(1-\alpha)\gamma(w_i)}{2\gamma(w_i)+\gamma(w_l)} \quad \mu_a(c) = \frac{(1-\alpha)\gamma(w_i)}{2\gamma(w_i)+\gamma(w_l)} \quad \mu_a(d) = \frac{(1-\alpha)\gamma(w_l)}{2\gamma(w_i)+\gamma(w_l)} \quad \mu_a(e) = 0 \quad \mu_a(f) = 0$$

$$\mu_b(a) = \frac{(1-\alpha)\gamma(w_i)}{2\gamma(w_i)+\gamma(w_l)} \quad \mu_b(b) = \alpha \quad \mu_b(c) = \frac{(1-\alpha)\gamma(w_i)}{2\gamma(w_i)+\gamma(w_l)} \quad \mu_b(d) = 0 \quad \mu_b(e) = \frac{(1-\alpha)\gamma(w_l)}{2\gamma(w_i)+\gamma(w_l)} \quad \mu_b(f) = 0$$

$$\mu_d(a) = (1-\alpha) \quad \mu_d(b) = 0 \quad \mu_d(c) = 0 \quad \mu_d(d) = \alpha \quad \mu_d(e) = 0 \quad \mu_d(f) = 0$$

Due to symmetry, we only need to consider $W(\mu_a, \mu_b)$ and $W(\mu_a, \mu_d)$ which represent the transportation cost for our two edge types w_i and w_l respectively.

For $W(\mu_a, \mu_b)$ take the transportation plan A and Kantorovich potential function $g(x)$ as

$$A(x,y) = \begin{cases} (1-\alpha)\frac{\gamma(w_i)}{2\gamma(w_i)+\gamma(w_l)} & \text{if } (x,y) = (a,a),(b,b),(c,c) \\ \alpha - (1-\alpha)\frac{\gamma(w_i)}{2\gamma(w_i)+\gamma(w_l)} & \text{if } (x,y) = (a,b) \\ (1-\alpha)\frac{\gamma(w_l)}{2\gamma(w_i)+\gamma(w_l)} & \text{if } (x,y) = (d,e) \\ 0 & \text{otherwise} \end{cases}$$

$$g(x) = \begin{cases} w_i & \text{if } x = a \\ w_i + w_l & \text{if } x = d \\ -w_l & \text{if } x = e \\ 0 & \text{otherwise} \end{cases}$$

It is straightforward to verify A and $g(x)$ are feasible and both yield the same transportation cost:

$W(\mu_a, \mu_b) = (\alpha - \frac{(1-\alpha)\gamma(w_i)}{2\gamma(w_i)+\gamma(w_l)})w_i + \frac{(1-\alpha)\gamma(w_l)}{2\gamma(w_i)+\gamma(w_l)}(w_i + 2w_l)$

Which results in the k-alpha curvature

$k^{\alpha}_{ab} = 1 - \frac{W(\mu_a,\mu_b)}{w_i} = (1-\alpha)\frac{3w_i\gamma(w_i)-2w_l\gamma(w_l)}{w_i(2\gamma(w_i)+\gamma(w_l))}$

Defining $k_{w_i} := k_{ab} = k_{ac} = k_{bc}$, we get

$k_{w_i} := k_{ab} = \lim_{\alpha\to 1} \frac{k^{\alpha}_{ab}}{1-\alpha} = \frac{3w_i\gamma(w_i)-2w_l\gamma(w_l)}{w_i(2\gamma(w_i)+\gamma(w_l))}$

For $W(\mu_a, \mu_d)$ take the transportation plan A and Kantorovich potential function $g(x)$ as

$$A(x,y) = \begin{cases} (1-\alpha) & \text{if } (x,y) = (a,a) \\ (2\alpha - 1) & \text{if } (x,y) = (a,d) \\ \frac{(1-\alpha)\gamma(w_i)}{2\gamma(w_i)+\gamma(w_l)} & \text{if } (x,y) = (b,d),(c,d) \\ \frac{(1-\alpha)\gamma(w_l)}{2\gamma(w_i)+\gamma(w_l)} & \text{if } (x,y) = (d,d) \\ 0 & \text{otherwise} \end{cases}$$

$$g(x) = \begin{cases} w_i & \text{if } x = a \\ w_i + w_l & \text{if } x = b,c,e,f \\ 0 & \text{otherwise} \end{cases}$$

It is straightforward to verify A and $g(x)$ are feasible and both yield the same transportation cost:

$W(\mu_a, \mu_d) = (2\alpha - 1)w_l + 2(w_l + w_i)\frac{(1-\alpha)\gamma(w_i)}{2\gamma(w_i)+\gamma(w_l)}$

Which results in the k-alpha curvature

$k^{\alpha}_{ad} = 1 - \frac{W(\mu_a,\mu_d)}{w_l} = 2(1-\alpha)\frac{w_l\gamma(w_i)+w_l\gamma(w_l)-w_i\gamma(w_i)}{w_l(2\gamma(w_i)+\gamma(w_l))}$

Defining $k_{w_l} := k_{ad} = k_{be} = k_{cf}$, we get

$k_{w_l} := k_{ad} = \lim_{\alpha\to 1} \frac{k^{\alpha}_{ad}}{1-\alpha} = \frac{2w_l\gamma(w_i)+2w_l\gamma(w_l)-2w_i\gamma(w_i)}{w_l(2\gamma(w_i)+\gamma(w_l))}$

We will choose $\gamma(x) = \frac{1}{x}$ as our gamma function. The unnormalized system becomes

$$\begin{cases} \frac{dw_l}{dt} = -\frac{2w_l}{2w_l+w_i}(w_l) \\ \frac{dw_i}{dt} = -\frac{w_l}{2w_l+w_i}(w_i) \end{cases}$$

Consider $r := \frac{w_i}{w_l}$. Then

$$\frac{dr}{dt} = \frac{w_i' w_l - w_i w_l'}{w_l^2} = \frac{w_i}{2w_l + w_i} = \frac{r}{2+r}$$

Where in the last step we substitute $w_i = rw_l$. Note that $r'(t) > 0$. Integrating, we get

$$\frac{2+r}{r} dr = dt \quad \to \quad \int (\frac{2}{r} + 1) dr = \int 1 dt \quad \to \quad 2\ln(r) + r = t + C$$

Where we don't need absolute value for ln since $r(t) > 0$. Since $r(0) = \frac{w_i(0)}{w_l(0)} = 1$, plugging in $t = 0$ and $r(0) = 1$ we get

$$2\ln(1) + 1 = 0 + C \quad \to \quad C = 1$$

Hence

$$2\ln(r) + r = t + 1$$

Which as $t \to \infty$ forces $r(t) \to \infty$.

We can express $\frac{dw_l}{dr}$ as the following

$$\frac{dw_l}{dt} = \frac{dw_l}{dr}\frac{dr}{dt} \quad \to \quad \frac{dw_l}{dr} = \frac{w_l'}{r'}$$

Since $r'(t) > 0$, this is valid. Substituting our expressions for w_l' and r' and $w_i = rw_l$ we get

$$\frac{dw_l}{dr} = \frac{w_l'}{r'} \quad \to \quad \frac{dw_l}{dr} = \frac{\frac{-2w_l}{2+r}}{\frac{r}{2+r}} = \frac{-2w_l}{r} \quad \to \quad \frac{1}{w_l} dw_l = -2\frac{1}{r} dr$$

Which we can integrate

$$\int \frac{1}{w_l} dw_l = \int -2\frac{1}{r} dr \quad \to \quad \ln w_l = -2\ln r + C \quad \to \quad w_l = Cr^{-2}$$

In the last step we exponentiate both sides and write e^C as C since it is some constant.

And since $r(0) = 1$ we get $C = w_l(0)$

Hence

$$w_l(t) = \frac{w_l(0)}{r(t)^2} \quad , \quad w_i(t) = r(t)w_l(t) = \frac{w_i(0)}{r(t)} = \frac{w_l(0)}{r(t)}$$

And since $r(t) \to \infty$ the unnormalized system results in $w_l(t) \to 0 \quad , \quad w_i(t) \to 0$. Therefore, the unnormalized flow collapses to all edge weights becoming 0. Note that w_l decreases to 0 at a faster rate than w_i.

Introducing normalization, the system becomes

$$\begin{cases} \frac{dw_l}{dt} = -\frac{2w_l}{2w_l+w_i}(w_l) + w_l \sum_h k_h w_h \\ \frac{dw_i}{dt} = -\frac{w_l}{2w_l+w_i}(w_i) + w_i \sum_h k_h w_h \end{cases}$$

Simplifying the normalization term we get

$$\sum_h k_h w_h = 3k_l w_l + 3k_i w_i = 3w_l$$

Consider the ratio $r = \frac{w_i}{w_l}$. Then

$$\frac{dr}{dt} = \frac{w_i' w_l - w_i w_l'}{w_l^2} = \frac{(w_i' + w_i \sum_h k_h w_h)w_l - (w_l' + w_l \sum_h k_h w_h)w_i}{w_l^2} = \frac{w_i}{2w_l + w_i} = \frac{r}{2+r}$$

To clarify confusion between unnormalized and normalized w_i' and w_l' terms, note that above we are writing the terms in the unnormalized version with the normalization terms directly included. The normalization terms cancel out and we are left with the same expression for r as in the unnormalized system. Hence we still have

$$2\ln(r) + r = t + 1$$

Which as $t \to \infty$ forces $r(t) \to \infty$.

We can express $\frac{dw_l}{dr}$ as the following

$$\frac{dw_l}{dt} = \frac{dw_l}{dr}\frac{dr}{dt} \quad \to \quad \frac{dw_l}{dr} = \frac{w_l'}{r'}$$

Since $w_i = rw_l \quad \to \quad 2w_l + w_i = w_l(2+r)$, substituting we get

$$\frac{dw_l}{dr} = \frac{w_l'}{r'} = \frac{-\frac{2w_l}{2w_l+w_i}(w_l) + w_l \sum_h k_h w_h}{\frac{r}{2+r}} = -\frac{2w_l}{r} + \frac{3(2+r)}{r} w_l^2$$

Let $y := \frac{1}{w_l}$. Then

$$\frac{dy}{dr} = -\frac{1}{w_l^2}\frac{dw_l}{dr} = -\frac{1}{w_l^2}\left(-\frac{2w_l}{r} + \frac{3(2+r)}{r} w_l^2\right) = \frac{2}{r}y - \frac{3(2+r)}{r}$$

Consider $x(r) := e^{(\int \frac{-2}{r}dr)} = r^{-2}$ and multiplying the above equation by r^{-2} on both sides,

$$\frac{dy}{dr}r^{-2} = \left(\frac{2}{r}y - \frac{3(2+r)}{r}\right)r^{-2} \quad \rightarrow \quad \frac{d}{dr}\frac{y}{r^2} = -\frac{6}{r^3} - \frac{3}{r^2}$$

Which we can integrate and get

$$\int \frac{d}{dr}\frac{y}{r^2}dr = \int \left(-\frac{6}{r^3} - \frac{3}{r^2}\right)dr \quad \rightarrow \quad \frac{y}{r^2} = 3r^{-2} + 3r^{-1} + C$$

multiplying both sides by r^2 and rewriting $y = \frac{1}{w_l}$ we get

$$\frac{1}{w_l} = 3 + 3r + Cr^2$$

And since $r(0) = 1$ we get $C = \frac{1}{w_l(0)} - 6 = 0$ since $w_l(0) = \frac{1}{6}$ by normalization and equal initial metric.

Hence

$$w_l(t) = \frac{1}{3(1+r(t))} \quad , \quad w_i(t) = r(t)w_l(t) = \frac{r(t)}{3(1+r(t))}$$

And since $r(t) \rightarrow \infty$, we get $w_l(t) \rightarrow 0$ and $w_i(t) \rightarrow \frac{1}{3}$. Therefore the normalized system results in the leaf edge weights going to 0 while the internal edge weights go to $\frac{1}{3}$. The curvature of the internal edges converge to $\frac{w_l}{2w_l+w_i} \rightarrow 0$. Since the leaf edges decay to weights 0 and the positive edge weight internal edges are all equal, we say the system is constant zero curvature by **Definition 6**.